

LangChain Retrievers: Complete Guide with Similarity Search

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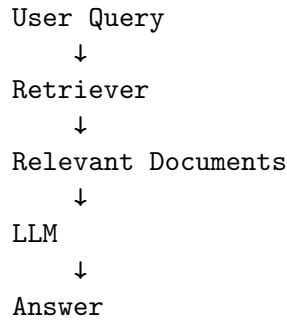
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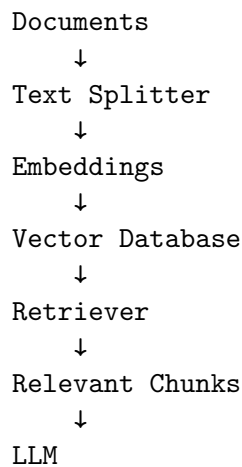
1 What is a Retriever?

A Retriever is responsible for fetching relevant documents from a knowledge base before sending them to the Large Language Model (LLM).

Workflow:



2 Retriever Workflow



3 Vector Store Retriever

The Vector Store Retriever uses embeddings to retrieve semantically similar documents.

3.1 Example

```
retriever = vector_store.as_retriever()

docs = retriever.invoke(
    "Explain Python frameworks"
)

for doc in docs:
    print(doc.page_content)
```

4 Similarity Search

Similarity Search compares the embedding of the query with document embeddings.

4.1 Process

Query
↓
Embedding Model
↓
Vector Similarity
↓
Top-K Documents

4.2 Similarity Search Example

```
results = vector_store.similarity_search(  
    "How do I build websites using Python?",  
    k=2  
)  
  
for doc in results:  
    print(doc.page_content)
```

5 Similarity Search with Scores

Returns both documents and similarity scores.

```
results = vector_store.similarity_search_with_score(  
    "Python frameworks",  
    k=2  
)  
  
for doc, score in results:  
    print(doc.page_content)  
    print(score)
```

6 Maximum Marginal Relevance (MMR)

MMR balances relevance and diversity.

Without MMR:

Django Tutorial
Django Setup
Django Installation

With MMR:

Django Tutorial
Flask Overview
FastAPI Guide

6.1 MMR Example

```

retriever = vector_store.as_retriever(
    search_type="mmr",
    search_kwargs={
        "k": 3,
        "fetch_k": 10
    }
)

```

7 Similarity Score Threshold Retriever

Returns documents above a threshold.

```

retriever = vector_store.as_retriever(
    search_type="similarity_score_threshold",
    search_kwargs={
        "score_threshold": 0.7,
        "k": 5
    }
)

```

8 BM25 Retriever

BM25 performs keyword-based retrieval.

8.1 Installation

```

pip install rank-bm25

```

8.2 Example

```

from langchain_community.retrievers import BM25Retriever

bm25 = BM25Retriever.from_documents(docs)

bm25.k = 2

results = bm25.invoke(
    "Python framework"
)

for doc in results:
    print(doc.page_content)

```

9 Similarity Search vs BM25

Feature	Similarity Search	BM25
Uses Embeddings	Yes	No
Semantic Understanding	Yes	No
Keyword Matching	No	Yes

Synonym Support	Yes	No
RAG Usage	Yes	Yes

10 Ensemble Retriever (Hybrid Search)

Combines keyword and vector retrieval.

```
from langchain.retrievers import EnsembleRetriever

vector_retriever = vector_store.as_retriever()

bm25_retriever = BM25Retriever.from_documents(docs)

ensemble = EnsembleRetriever(
    retrievers=[
        bm25_retriever,
        vector_retriever
    ],
    weights=[0.5, 0.5]
)

results = ensemble.invoke(
    "Python web frameworks"
)
```

11 MultiQuery Retriever

Generates multiple query variations.

Example:

What is RAG?

↓

Explain Retrieval-Augmented Generation

Describe how RAG works

What are RAG systems?

11.1 Example

```
from langchain.retrievers.multi_query import MultiQueryRetriever

multi = MultiQueryRetriever.from_llm(
    retriever=vector_retriever,
    llm=llm
)
```

12 Parent Document Retriever

Stores small chunks but returns parent documents.

PDF

↓

Small Chunks Retrieved

↓
Parent Document Returned

13 Contextual Compression Retriever

Filters irrelevant content.

Retrieved Documents
↓
Compression
↓
Relevant Information
↓
LLM

14 Self Query Retriever

Uses metadata automatically.

Example Metadata:

```
metadata = {  
    "year": 2025,  
    "author": "Aftab"  
}
```

Query:

Show AI articles after 2024.

Converted to:

year > 2024

15 Time Weighted Retriever

Prioritizes recent documents.

Useful for:

- Chat Memory
- News Systems
- Logs

16 Merger Retriever

Combines outputs from multiple retrievers.

Example:

FAISS
+
Chroma
+
BM25

17 External Retrievers

- Wikipedia Retriever
- Tavily Retriever
- Amazon Kendra Retriever
- Arxiv Retriever

18 Retriever Comparison

Retriever	Embeddings	Metadata	Best Use
VectorStore	Yes	Limited	Standard RAG
BM25	No	No	Exact Match
Ensemble	Yes	Limited	Hybrid Search
MultiQuery	Yes	No	Better Recall
ParentDocument	Yes	No	Large PDFs
Compression	Yes	No	Reduce Tokens
SelfQuery	Yes	Yes	Metadata Filtering
TimeWeighted	Optional	Time	Memory Systems

19 Production RAG Stack (2026)

Embeddings (BGE/Gemini)

↓

Qdrant or Pinecone

↓

Similarity + BM25

↓

Ensemble Retriever

↓

Reranker

↓

LLM

↓

Final Answer

20 Interview Questions

1. What is a Retriever?
2. What is Similarity Search?
3. Difference between BM25 and Vector Search?
4. What is Ensemble Retriever?
5. Explain MMR.
6. What is MultiQuery Retriever?
7. What is Parent Document Retriever?

8. What is Contextual Compression?
9. What is Self Query Retriever?
10. Which Retriever would you use in Production?